

compromise with a 3.71 percent initial spending rate, and the initial spending rate for the moving average endowment formula is identical to that for the fixed-percentage rule.

Bengen's floor-and-ceiling rule allows the initial withdrawal rate to increase to 3.20 percent and helps keep downside spending at a level close to the baseline amount. Vanguard's floor-and-ceiling rule again behaves closer to a fixed-percentage strategy, though it preserves wealth at a stronger pace. Next, the Kitces ratcheting rule has the lowest initial spending rate of 2.69 percent, but it allows for significant upside spending potential while keeping the downside risk similar to the baseline.

Both the Guyton and Klinger decision rules and the Zolt rule allow initial spending to rise about 4 percent to 4.47 percent and 4.64 percent, respectively, though their subsequent evolution varies as before. Finally, the modified RMD rule is calibrated to the PAY Rule by using a 1.11 multiplication factor for the RMD rates, and it again leads to large upside spending potential and the most downside spending volatility.

Exhibit 6.21

Sustainable Spending Rates from an Investment Portfolio over 30 Years

PAY Rule: Allow for a 10% Chance That Real Wealth Has Fallen Below \$15,000 by Year 30

*For a 65-Year-Old Couple, Using a 50/50 Portfolio of Stocks and Bonds
1,000 Monte Carlo Simulations Based on Current Market Conditions
Retirement Date Wealth Level = \$100,000*

Spending Strategy			Real	Real	Real	Real	Change	Year 30	
			Spending	Spending	Spending	Remaining	in Real	Spending	
	Initial	Percentile	in 10	in 20	in 30	Wealth	Spending	Relative	Downside
	Spending	of	Years	Years	Years	After 30	at Year	to	Spending
	Rate	Distribution				Years	30	Baseline	Volatility
Constant Inflation-		90th	\$2,780	\$2,780	\$2,780	\$360,480	0%	0%	0%
Adjusted Spending	2.78%	50th	\$2,780	\$2,780	\$2,780	\$109,950	0%	0%	0%
(BASELINE)		10th	\$2,780	\$2,780	\$2,780	\$14,880	0%	0%	0%